

Project Report Format

Leave Approval System

1. INTRODUCTION

1.1 Project Overview

The **Leave Approval System** is a ServiceNow-based application developed to automate and streamline the leave management process within an organization. The system enables employees to submit leave requests by providing details such as leave type, start date, end date, and reason for leave. These requests are then routed to the respective managers for review and approval through an automated workflow. Managers can approve or reject requests and provide comments, ensuring a transparent and efficient approval process. The application maintains centralized records of all leave requests, making it easier to track and manage leave information. By reducing manual paperwork and improving communication between employees and managers, the system enhances operational efficiency, accuracy, and productivity within the organization.

1.2 Purpose

The purpose of this project is to develop an automated **Leave Approval System** that simplifies the process of managing employee leave requests. The system provides a centralized platform for employees to submit leave applications and for managers to review, approve, or reject them efficiently. It aims to reduce manual effort, minimize errors in leave tracking, improve transparency in the approval process, and maintain accurate records of employee leave data. The project also enhances organizational productivity by ensuring faster communication and streamlined workflow management.

2. IDEATION PHASE

2.1 Problem Statement

In many organizations, the leave management process is still handled manually through paper-based forms, emails, or spreadsheets. These traditional methods often lead to delays in approval, data inconsistencies, misplaced records, and a lack of transparency in tracking leave requests. Employees may find it difficult to know the status of their applications, while managers spend significant time reviewing and managing requests manually. Additionally, maintaining accurate leave records and generating reports becomes challenging as the organization grows.

To address these issues, there is a need for an automated **Leave Approval System** that provides a centralized platform for submitting, reviewing, and managing leave requests. The system should automate approval workflows, maintain accurate records, improve communication between employees and managers, and ensure efficient tracking of leave statuses. This solution will reduce administrative effort, minimize errors, enhance transparency, and improve overall organizational productivity.

2.2 Empathy Map Canvas

The empathy map was created to understand the expectations, challenges, and experiences of employees using the Leave Approval System.

User Persona

Employee requesting leave through the ServiceNow Leave Approval System Think & Feel

- I hope my leave gets approved quickly.
- I want a simple and easy leave application process.
- I don't want to make mistakes in dates.

Hear

- Company leave policy and procedures.
- Manager's instructions and feedback.
- HR announcements and updates.
- Email notifications (approval/rejection).

See

- Leave Approval application in ServiceNow.
- Leave types (CL, EL, SL, PL, etc.).
- Start Date and End Date fields.
- Status of my request (Pending, Approved, Rejected).

Say & Do

- Select leave type.
- Enter Start Date, End Date, and reason.
- Submit leave request.
- Check the status of my leave request.
- Wait for manager approval.

Pain Points

- If the End Date is earlier than the Start Date, it shows an error.
- Approval takes too long.
- Not sure if the manager saw my request.
- No clarity on leave status sometimes.

Gains

- Automatic manager approval using Flow Designer.
- Validation prevents End Date from being earlier than the Start Date.
- Instant email notifications for approval/rejection.
- Transparent leave status tracking



2.3 Brainstorming

The project team conducted brainstorming sessions to identify an effective solution for automating the leave approval process. Several ideas were discussed and evaluated based on feasibility, implementation complexity, user experience, and business value.

Brainstormed Ideas

- Develop an online leave request form.
- Automate manager approval using Flow Designer.
- Send approval and rejection email notifications.
- Validate leave dates using Client Scripts.
- Store leave requests in a centralized database.
- Display leave approval status to employees.
- Generate leave reports for administrators.
- Maintain secure digital leave records.

Idea Prioritization

- After evaluating all ideas, the following features were selected for implementation:
- Employee Leave Request Form
- Flow Designer Approval Workflow
- Manager Approval Process
- Client Script Date Validation
- Automatic Email Notification
- Leave Status Tracking
- Centralized Leave Request Database

These features were selected because they provide the highest value to users while being practical to implement using the ServiceNow Zurich Platform.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Customer Map Journey

The Customer Journey Map illustrates the complete experience of an employee while using the Leave Approval System, from submitting a leave request to receiving the final approval or rejection notification. It helps identify customer interactions, expectations, challenges, and opportunities for improving the leave management process.

Scenario

An employee submits a leave request through the ServiceNow Leave Approval System. The system validates the leave details, routes the request to the reporting manager for approval, updates the leave status, and sends an email notification to the employee.

Customer Journey Stages

Awareness

- Employee becomes aware of the Leave Approval System through the organization or HR department.

Access

- Employee logs into the ServiceNow portal and opens the Leave Request application.

Submission

- Employee enters leave details, including employee name, email, leave type, dates, manager, and reason for leave.

Validation

- The system validates the entered details using Client Scripts to prevent incorrect submissions.

Approval

- Flow Designer automatically routes the leave request to the reporting manager for approval.

Status Update

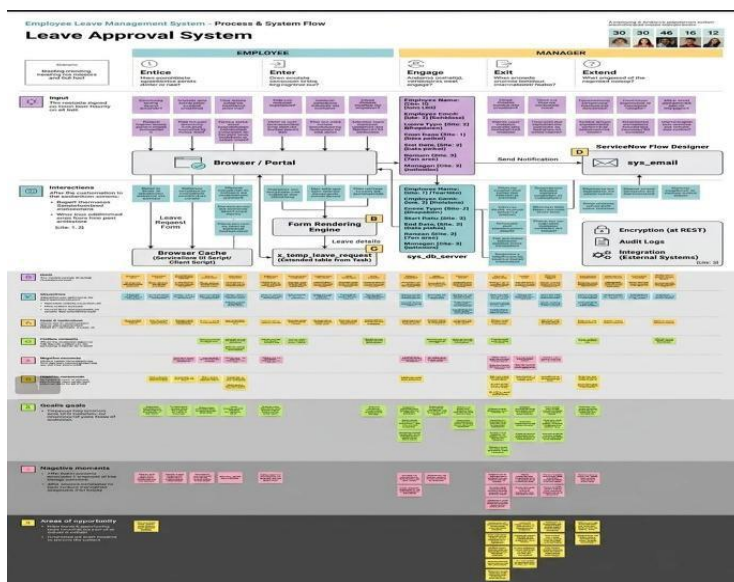
- The manager approves or rejects the leave request, and the system updates the request status.

Notification

The employee receives an automated email informing them whether the leave request has been approved or rejected.

Follow-up

- The employee can view the updated leave request status and refer to previous leave records whenever required.



3.2 Solution Requirement

Functional Requirements

The Leave Approval System provides the following functional requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Leave Application	Employee submits leave request by selecting Leave Type, Start Date, End Date, and Reason.
FR-2	Date Validation	System validates that the End Date is not earlier than the Start Date before submission.
FR-3	Manager Approval	Flow Designer sends the leave request to the assigned manager for approval or rejection.
FR-4	Leave Status Update	System updates the leave request status to Pending, Approved, or Rejected.

FR-5	Email Notifications	System sends email notifications to employees when the leave request is approved or rejected.
FR-6	Leave Record Management	Administrator can view and manage leave requests and records.

Non Functional Requirements

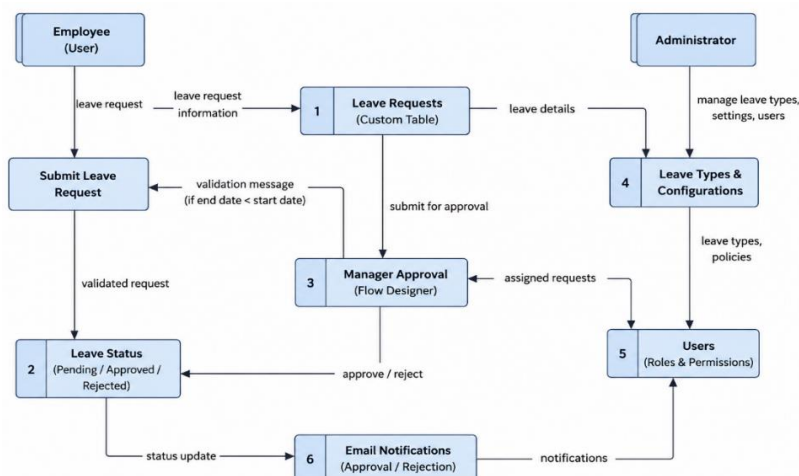
Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple and user-friendly interface for submitting and managing leave requests.
NFR-2	Security	Only authorized users (Employees, Managers, and Administrators) can access their permitted functions using role-based access control.
NFR-3	Reliability	The system should process leave requests accurately and maintain correct approval status without data loss.
NFR-4	Performance	Leave requests, approvals, and notifications should be processed within a few seconds under normal usage.
NFR-5	Availability	The Leave Approval System should be available whenever users need to submit or approve leave requests.
NFR-6	Scalability	The system should support an increasing number of employees and leave requests without affecting performance.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of how data moves through a system. It illustrates the flow of information between external entities, processes, and data stores. In the **Leave Approval System**, the DFD shows how employees submit leave requests, how managers review and approve or reject those requests, and how the system maintains leave records. The diagram helps in understanding the overall workflow and interactions within the system.

The Leave Approval System begins when an employee submits a leave request containing details such as leave type, dates, and reason. The system processes the request and forwards it to the respective manager for review. The manager then approves or rejects the request and may provide comments. The system updates the leave status and stores the information in the leave records database. Finally, the employee is notified of the decision. This automated flow ensures accuracy, transparency, and efficient management of employee leave requests.



3.4 Technology Stack

The Leave Approval System is developed using the ServiceNow Zurich Platform and utilizes various platform components to automate leave management.

Component	Technology Used
Platform	ServiceNow Zurich
Programming Language	JavaScript (Client Script)
Workflow Automation	Flow Designer
Database	ServiceNow Custom Table
Validation	Client Script
Approval Process	ServiceNow Approval Engine
Notifications	ServiceNow Email Notification
User Interface	ServiceNow Form UI

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed **Leave Approval System** effectively addresses the challenges associated with traditional leave management processes. Organizations often face issues such as delayed approvals, manual paperwork, lack of transparency, and difficulty in maintaining accurate leave records. These problems can lead to inefficiencies, communication gaps, and increased administrative workload.

To overcome these challenges, the Leave Approval System provides an automated and centralized platform for managing leave requests. Employees can easily submit leave applications, while managers can review, approve, or reject requests through a streamlined workflow. The system maintains accurate records, provides real-time status tracking, and improves communication between employees and managers. By automating the entire process, the solution reduces manual effort, minimizes errors, enhances transparency, and improves overall organizational efficiency.

4.2 Proposed Solution

The proposed solution is to develop a **Leave Approval System** using the ServiceNow platform to automate and simplify the leave management process. The system provides a centralized platform where employees can submit leave requests by entering details such as leave type, start date, end date, and reason for leave. Once submitted, the request is automatically routed to the respective manager for review and approval.

Managers can approve or reject leave requests and provide comments through an automated workflow. The system updates the leave status in real time and maintains a centralized database of all leave records for future reference. Notifications are sent to employees regarding the status of their requests, ensuring clear communication and transparency. This solution reduces manual effort, minimizes errors, improves record management, and enhances overall organizational efficiency.

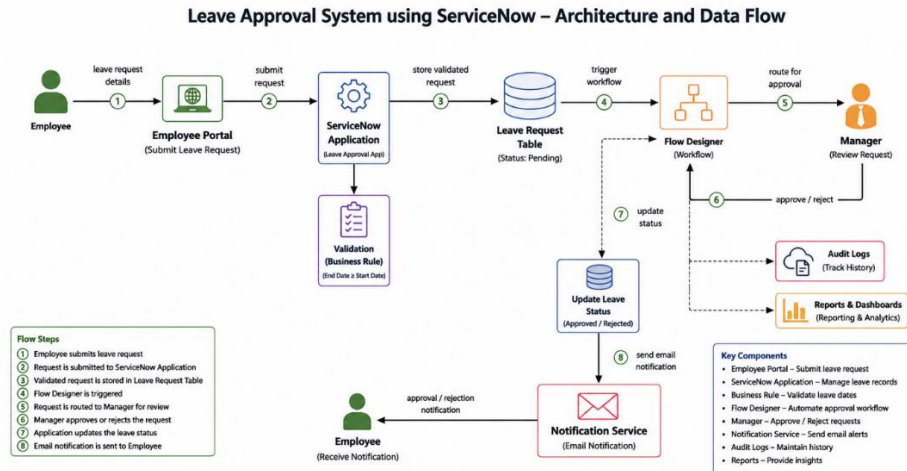
4.3 Solution Architecture

The **Leave Approval System** is designed using the ServiceNow platform to automate the leave management process. The architecture consists of three main components: the Employee Interface, the Leave Approval Processing Module, and the Manager Approval Interface. Employees submit leave requests through a user-friendly form, which captures details such as leave type, start date, end date, and reason for leave.

Components of the Solution Architecture

- Employee
- Employee Portal
- ServiceNow Application
- Leave Request Form
- Client Script Validation

- Leave Request Table
- Flow Designer
- Manager Approval
- Leave Status Update
- Email Notification
- Reports & Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The development of the **Leave Approval System** was planned in a structured manner to ensure successful implementation. The project began with requirement gathering and analysis, where the challenges of the existing leave management process were identified. Based on these requirements, the system design was prepared, including database tables, fields, forms, and approval workflows.

The implementation phase involved creating the Leave Approval table, configuring fields, designing forms, and developing automated approval workflows in ServiceNow. After development, the system was tested to verify the accuracy of leave request submission, approval processing, and status tracking. Finally, the project was deployed and documented to ensure smooth usage and future maintenance. This systematic approach helped in delivering an efficient and reliable leave management solution.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Custom Application	USN-1	As an employee, I can submit a leave request by entering leave type, start date, end date, and reason.	5	High
Sprint-1	Date Validation	USN-2	As a user, I want the system to prevent the end date from being earlier than the start date.	3	High
Sprint-2	Manager Approval	USN-3	As a manager, I can approve or reject leave requests using Flow Designer.	5	High
Sprint-2	Status Update	USN-4	As an employee, I can view the status of my leave request	3	Medium

			(Pending, Approved, Rejected).		
Sprint-3	Email Notification	USN-5	As an employee, I receive email notifications when my leave request is approved or rejected.	4	High
Sprint-3	Leave Records	USN-6	As an administrator, I can view and manage all leave requests.	3	Medium
Sprint-4	Testing	USN-7	As an administrator, I can test the leave approval workflow to ensure it works correctly.	4	High

Velocity:

Total Story Points = 30

Sprint Duration = 20 Days (4 sprints × 5 days)

Average Velocity = Total Story Points ÷ Total Days

Average Velocity = 30 ÷ 20 = 1.5 Story Points per Day

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to evaluate the efficiency and responsiveness of the **Leave Approval System** under normal operating conditions. The system was tested by submitting multiple leave requests, processing approvals, and updating request statuses to ensure smooth functionality. The results showed that the system could handle user requests efficiently without significant delays or errors.

The testing also verified the accuracy of data storage, workflow execution, and notification delivery. All leave requests were processed successfully, and approval actions were reflected in real time. The system demonstrated reliable performance, quick response times, and consistent operation, confirming its suitability for managing employee leave requests in an organizational environment.

7. RESULTS

7.1 Output Screenshots

The screenshot shows the ServiceNow 'Leave Approval - Create' form. The header includes the ServiceNow logo and navigation links. The form fields are as follows:

- Employee:** Angellique Schermerhorn
- Manager:** Abel Tuter
- Leave Type:** Earned Leave (dropdown menu)
- Start Date:** 2026-07-02
- End Date:** 2026-07-06
- Reason:** vacation
- Status:** Pending (dropdown menu)
- Manager Comments:** (empty text area)

A 'Submit' button is located at the bottom left of the form.

Impersonate user

Select a user

Search for a user

RECENT IMPERSONATIONS

Abel Tuter
abel.tuter

Abraham Lincoln
abraham.lincoln

ITIL User
itil

Cancel

Impersonate user

servicenow

AllFavoritesHistoryWorkspacesAdmin

Approvals

Application scope: Global
Update set: Default [Global]

Approvals

State

Search

Actions on selected rows...

All > Sys ID = NULL .or. Approver = Abel Tuter

State	Approver	Comments	Approval for	Created
Approved	Abel Tuter		(empty)	2026-06-28 09:34:22
Approved	Abel Tuter		(empty)	2026-06-28 07:38:56
Rejected	Abel Tuter		(empty)	2026-06-28 08:43:46
Approved	Abel Tuter		(empty)	2026-06-28 06:20:15
Approved	Abel Tuter		(empty)	2026-06-28 08:10:02
Approved	Abel Tuter		(empty)	2026-06-28 07:00:21
Approved	Abel Tuter		(empty)	2026-06-28 04:42:45
Approved	Abel Tuter		(empty)	2026-06-28 08:09:37
Requested	Abel Tuter		(empty)	2026-07-01 07:35:03
Rejected	Abel Tuter		(empty)	2026-06-28 07:48:31
Approved	Abel Tuter		(empty)	2026-06-28 07:57:38
Approved	Abel Tuter		(empty)	2026-06-28 07:55:23
Requested	Abel Tuter		(empty)	2026-07-01 07:32:23

Show Matching

Filter Out

Copy URL to Clipboard

Copy sys_id

Assign Tag

Approve

Reject

1 to 13 of 13



servicenow All Favorites History Workspaces Emails View: Outlook Application scope: Leave Approval System Update set: Default (Global)

Emails View: Outlook Created Search Actions on selected rows... New

All > Mailbox > Outlook > Created on Today

	Created	Recipients	Subject	State
☐	2026-07-01 07:36:12	alleen.mottern@example.com	Leave Request Approved	Ready
☐	2026-07-01 07:32:59	abraham.lincoln@example.com	Leave Request Approved	Ready
☐	2026-07-01 06:00:09	fred.luddy@example.com	On-Call Schedule: Hardware	Ready
☐	2026-07-01 01:00:09	alleen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
☐	2026-07-01 01:00:09	alleen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
☐	2026-07-01 01:00:09	alleen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
☐	2026-07-01 00:00:09	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready

8. ADVANTAGES & DISADVANTAGES

Advantages of Leave Approval System

- Automates the leave request and approval process, reducing manual work.
- Provides faster approval and notification to employees.
- Maintains accurate and centralized leave records.
- Improves transparency by allowing employees to track leave status.
- Reduces paperwork and minimizes data entry errors.

Disadvantages of Leave Approval System

- Depends on system availability and internet connectivity.
- Initial setup and configuration may require time and technical expertise.
- System errors or workflow issues can delay approvals.
- Users may require training to use the application effectively.

- Regular maintenance and updates are needed to ensure smooth operation.

9. CONCLUSION

The Leave Approval System is an efficient solution for managing employee leave requests and approvals in an organization. It automates the leave management process, reduces manual effort, improves accuracy, and ensures transparency between employees and managers. By maintaining centralized records and streamlining approval workflows, the system enhances productivity and provides a reliable method for tracking and managing employee leave.

10. FUTURE SCOPE

- Integration with attendance and payroll management systems.
- Email and SMS notifications for leave requests and approvals.
- Development of a mobile application for easy access.
- Automated leave balance calculation and tracking.
- Advanced reporting and analytics dashboards.
- Support for multiple departments and organizational units.
- Enhanced security through role-based access control.
- Integration with calendar applications for leave scheduling.
- Employee self-service portal for leave management.